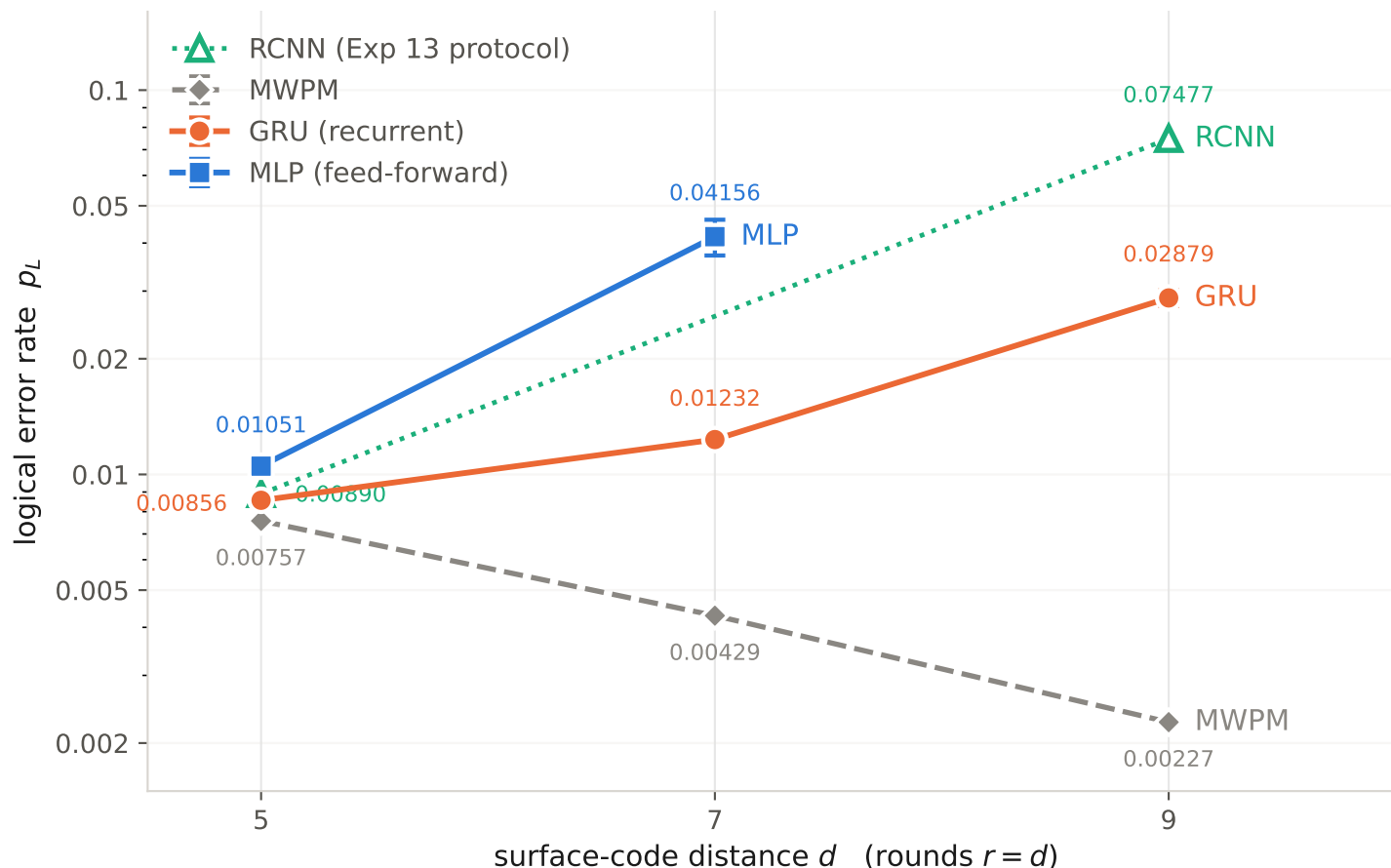


Experiment 16 — recurrence becomes more important as distance grows



MWPM improves with distance while both learned decoders get worse, and the feed-forward model degrades fastest. GRU and MLP are parameter-matched within each distance (69,441 vs 69,389 at $d=5$; 79,521 vs 79,841 at $d=7$) and share one protocol: 10M training shots, batch 10,000, 200 fixed epochs, minimum-validation-loss checkpoint, $p=0.004$, scored on the same 1.8M shots. Error bars are the standard deviation over seeds; MWPM carries its binomial error.

The RCNN line is carried over from Experiment 13 for reference and follows a different protocol: batch 5,000, epoch-indexed LR schedule with early stopping, one seed, and the EAF pools. It is not parameter-matched to the students and has no $d=7$ point.